Course Outline

Sem		Course Name	Th	Tut	Pra	Hrs	Cr
Sem I	HSC	Humanities – 1	2	0	0	2	2
	BSC	Mathematics – 1	3	1	0	4	4
	BSC	Physics	3	0	0	3	3
	BSC	Physics – Laboratory	0	0	2	2	1
	BSC	Environment Science	2	0	0	2	2
	ESC	Workshop Practices	0	0	2	2	1
	BSC	Biology for Engineers	2	0	0	2	2
	ESC	Computer Programming -I	1	0	0	1	1
	ESC	Computer Programming -I Laboratory	0	0	2	2	1
	HSC	Universal Human Values	1	0	0	1	1
	HSC	Indian Knowledge System	2	0	0	2	2
			16	**1**	**5**	**23**	**20**
Sem II	HSC	Humanities – 2	2	0	0	2	2
	BSC	Mathematics – 2	3	1	0	4	4
	BSC	Chemistry	3	0	0	3	3
	BSC	Chemistry – Laboratory	0	0	2	2	1
	ESC	Introduction to Electrical Engineering	3	0	0	3	3
	ESC	Introduction to Electrical Engineering-Laboratory	0	0	2	2	1
	ESC	Engineering Graphics	0	0	4	4	2
	HSC	Yoga, Health & Hygiene OR NCC/NSS	0	0	2	2	1
	HSC	Organizational Behaviour	1	0	0	1	1
	ESC	Computer Programming -II	1	0	0	1	1
	ESC	Computer Programming -II Laboratory	0	0	2	2	1
			13	**1**	**12**	**26**	**20**
Sem III	Pro	Civic and Social Service Internship	0	0	0	0	1
	ESC	Introduction to Artificial Intelligence (Department Specific)	3	0	0	3	3
	PC	Digital Circuits	3	0	0	3	3
	PC	Digital Circuits Laboratory	0	0	2	2	1
	PC	Data Structures and Algorithms	3	0	0	3	3
	PC	Data Structures and Algorithms Laboratory	0	0	2	2	1
	PC	Fundamentals of ICT	2	0	0	2	2
	PC	Electronics Devices and Circuits	2	0	0	2	2
	BSC	Discrete Mathematical Structures	3	1	0	4	4
			16	**1**	**4**	**21**	**20**
Sem IV	ESC	Industry 4.0 (Department Specific)	2	0	0	2	2
	ESC	Industry 4.0 – Laboratory (Department Specific)	0	0	2	2	1
	OE	Open Elective 1 (From Other School)	3	0	0	3	3
	PC	Database Management Systems	3	0	0	3	3
	PC	Database Management Systems Laboratory	0	0	2	2	1
	PC	Principles of Programming Languages	3	0	0	3	3
	PC	Principles of Programming Languages Laboratory	0	0	2	2	1

3rd Semester

24ICxxxT					Digital Circuits					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hrs/Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
3	0	0	3	3	25	50	25	0	0	100

COURSE OBJECTIVES

- To learn the fundamentals of digital logic circuits and digital number systems.
- To be able to analyze and design digital combinational-logic and sequential-logic circuits.
- To impart to student the concepts of sequential circuits, enabling them to analyze sequential systems in terms of state machines.

UNIT-1: NUMBER SYSTEMS AND LOGIC GATES	**8 Hrs.**
Digital systems; positional number system; binary, octal and hexadecimal representation of integer/fixed-point numbers; Methods of base conversions; complements; Representation of signed numbers; signed/unsigned binary, octal and hexadecimal arithmetic; Binary coded decimal codes; Gray codes; Excess-3 codes, Logic Gates, and Truth Table; Error detection and correction codes - parity check codes and Hamming codes.	
UNIT-2: BOOLEAN ALGEBRA AND COMBINATIONAL CIRCUITS	**14 Hrs.**
Boolean Algebra & Simplification of Boolean Algebra: Basic postulates and fundamental theorems of Boolean algebra; Representation of logic functions: canonical and standard forms; Simplification of Boolean functions: K-map and Quine-McCluskey tabular methods.	
Definition of combinational circuits; design procedure; Decoders; encoders; multiplexers; demultiplexers and their applications; Parity generator and checker circuits; comparators; Arithmetic modules- ripple carry adder, subtractor, carry look ahead adder; basic multiplier, BCD Adder; Shannon expansion theorem.	
UNIT-3: SEQUENTIAL CIRCUITS	**14 Hrs**.
Definition of sequential circuits; Latches and flip-flops: SR-latch; D-latch; D flip-flop; JK flip-flop; T flip-flop.	
Synchronous sequential circuit; Moore and Mealy Finite State Machines (FSM): state diagram, state table, state assignment; Analysis and implementation of FSM using D flip-flops, JK flip-flops and T flip-flops; Implementation of FSM using Decoders, OR gates, Multiplexors and ROM.	
Various Registers and counters: asynchronous counters, synchronous counters; ring counter, Johnson counter.	
UNIT 4: INTRODUCTION TO MEMORY AND PROGRAMMABLE LOGIC CIRCUITS	**6 Hrs**.
Introduction, Random access memory, read only memory, programmable logic array, programmable array logic, sequential programmable devices, FPGA.	**Max. Hrs.: 42**

COURSE OUTCOMES

On completion of the course, the student will be able to

CO1: Study the basics of number systems, Boolean algebra and standard digital logic gates.
CO2: Understand the knowledge of Boolean Algebra and simplification of Boolean expressions to deduce optimal digital networks.
CO3: Apply the knowledge to synthesis combinational circuits.
CO4: Analyse the operation and logic circuits of basic storage elements FFs.
CO5: Implement SSI and MSI digital networks given a state diagram based on Mealy and Moore configurations.
CO6: Design and examine the SSI, MSI and Programmable combinational networks.

TEXT/REFERENCE BOOKS

1. Digital Design - M. Morris Mano and Michael Ciletti, 5 th edition, Pearson.
2. Switching Theory and Logic Design - A. Anand Kumar, PHI, 2nd Edition.
3. Fundamentals of Digital Logic with Verilog Design - Stephen Brown and Zvonko Vranesic Tata McGraw Hill.
4. Switching and Finite Automata Theory - Zvi Kohavi & Niraj K. Jha, 3rd Edition, Cambridge.
5. Fundamentals of Logic Design - Charles H. Roth, Cengage LEanring, 5th, Edition.
6. Introduction to Switching Theory and Logic Design – Fredriac J. Hill, Gerald R. Peterson, 3rd Ed, John Wiley & Sons Inc.
7. Digital Fundamentals – A Systems Approach – Thomas L. Floyd, Pearson.

24ICxxxP					Digital Circuits Lab					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hrs/Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
0	0	2	1	2	--	--	--	25	25	50

COURSE OBJECTIVES:

- To learn the fundamentals of digital logic design and digital circuits.
- To be able to analyze and design digital combinational-logic and sequential-logic circuits.
- To be able to implement digital combinational-logic and sequential-logic circuits.

List of Experiments

1. Study of Different types of logic gates and implement them using NAND and NOR gates.
2. Design a simple combinational circuit with four variables and obtain minimal expression and verify the truth table.
3. Study of Half and full adder circuits.
4. Study of Binary and BCD parallel adder circuits.
5. Study of Decoder and Encoder circuits.
6. Study of Multiplexer and De-multiplexer circuits.
7. Study of Different types of flip flops.
8. Study of Asynchronous and synchronous binary counters.
9. Study of Various types of registers.
10. Design a BCD to 7-segment display decoder circuit and interface with a 7-sgment display.

COURSE OUTCOMES

On completion of the course, the student will be able to:

CO1 - Remember digital logic circuit functionalities
CO2 - Understand the design methodologies to utilize them practically.
CO3 - Apply design methods to realize digital circuits practically.
CO4 - Analyze functionalities of digital circuits.
CO5 - Evaluate and report performance of digital circuits.
CO6 - Design and Test elementary digital combinational-logic and sequential-logic circuits.

TEXT/REFERENCE BOOKS

- M. Morris Mano and Michael Ciletti, “Digital Design: With a Introduction to the Verilog HDL”, 5th edition,Pearson.

<Course Code>					Data Structures and Algorithms					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hrs./Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
3	1	0	4	4	25	50	25	--	--	100

COURSE OBJECTIVES

1. To impart the basic concepts of data structures and algorithms
2. To explain different types of data structures such as stacks, queues, lists, trees and graphs
3. To understanding about writing algorithms and step by step approach in solving problems with the help of fundamental data structures

UNIT I: INTRODUCTION TO DATA STRUCTURE	**12 Hrs.**
Introduction: Basic Terminologies: Elementary Data Organizations, Data Structure Operations: insertion, deletion, traversal etc.; Analysis of an Algorithm, Asymptotic Notations, Time-Space trade off Searching: Linear Search and Binary Search Techniques and their complexity analysis. Stacks and Queues, Applications of Stacks: Expression Conversion and evaluation corresponding algorithms and complexity analysis.	
UNIT II: LINEAR DATA STRUCTURE	**10 Hrs.**
Array, sparse matrix, and its representation. Stacks, Operations, Applications,Polish Expression, Reverse Polish Expression, and Their Compilation. Queue: Representation, Operations, Circular Queue, Priority Queue, Array Representation of Priority Queue, Double-Ended Queue, Applications of Queue Linked List: Singly & Doubly Linked list, Circular linked list, Linked implementation of Stack, Linked implementation of Queue, Applications of linked list.	
UNIT III: NON-LINEAR DATA STRUCTURE	**10 Hrs.**
Trees: Basic Tree Terminologies, Different types of Trees: Binary Tree, Threaded Binary Tree, Binary Search Tree, AVL Tree; Tree operations on each of the trees and their algorithms with complexity analysis. Applications of Binary Trees. B Tree, B+ Tree: definitions, algorithms and analysis. Graph: Basic Terminologies and Representations, Graph search and traversal algorithms and complexity analysis.	
UNIT IV: SORTING AND SEARCHING AND HASHING	**10 Hrs.**
Sorting and Hashing: Objective and properties of different sorting algorithms: Selection Sort, Bubble Sort, Insertion Sort, Quick Sort, Merge Sort, Heap Sort	
	TOTAL HOURS: 42 Hrs.

COURSE OUTCOMES

On completion of the course, student will be able to:

CO1	:	Define the basic concepts of data structures to represent data items in real world.
CO2	:	Understand the concepts of algorithm evaluation and find time and space complexities.
CO3	:	Apply the linear data structure, such as stacks, queues, and linked lists, for various applications.
CO4	:	Analyse the basic operations on binary trees.
CO5	:	Evaluate traversal techniques of graphs for certain applications.
CO6	:	Design and develop the application of algorithms for sorting and pattern matching.

TEXT/REFERENCE BOOKS

1. Jean-Paul Tremblay & Paul G "**An Introduction to Data Structures with Applications**", Tata McGraw Hill.
2. Ten Baum "**Data Structures using C & C++**", Prenctice-Hall International.
3. Thomas H. Cormen, Charles E. Leiserson, Ronald L Rivest, Clifford Stein "**Introduction to Algorithms**", Third Edition, PHI Learning.
4. Sanjoy DasGupta, C. H. Papadimitriou, Umesh Vazirani "**Algorithms**", First Edition, Tata McGraw Hill, 2006. Data Structures:
5. E.Horowitz, S.Sahni and Susan AndersonFreed "**Fundamentals of Data structures in C**", 2nd Edition, Universities Press

<Course Code>					Data Structures and Algorithms Laboratory					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hrs/Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
0	0	2	1	2	--	--	--	50	50	100

COURSE OBJECTIVES

1. To implement linear and non-linear data structures.
2. To understand the different operations of search trees.
3. To implement graph traversal algorithms.
4. To get familiarized to sorting and searching algorithms.

LIST OF EXPERIMENTS	
1	Write a C program to perform matrix multiplication using array.
2	(a) Write a C program to create a stack using an array and perform: (i) push operation (ii) pop operation (b) Write a C program to create a queue and perform: i) Push ii) pop iii) Traversal
3	Write a C program that uses Stack operations to perform the following: i) Converting infix expression into postfix expression ii) Evaluating the postfix expression
4	Write a C program that uses functions to perform the following operations on a Single linked list: i) Creation ii) Insertion iii) Deletion iv) Traversal in both ways
5	Write a C program that uses functions to perform the following operations on Double linked list: i) Creation ii) Insertion iii) Deletion
6	Write a C program that uses functions to perform the following operations on Binary Tree: i) Creation ii) Insertion iii) Deletion
7	Write C programs that use both recursive and nonrecursive functions to perform the Linear search operation.
8	Write C program that use both recursive and non-recursive functions to perform the Binary search operation.
9	Write a C program that implement Bubble Sort method to sort a given list of integers in descending order.
10	Write a C program that implement Quick Sort method to sort a given list of integers in ascending order.

COURSE OUTCOMES

On completion of the course, student will be able to:

CO1	:	Understand the appropriate data structures as applied to the specified problem definition.
CO2	:	Apply operations like searching, insertion, and deletion, traversing mechanism, etc. on various data structures.
CO3	:	Analyze Linear and Non-Linear data structures.
CO4	:	Evaluate the appropriate sorting/searching technique for the problem.
CO5	:	Design advanced data structure using Non-linear data structure.
CO6	:	Determine and analyze the complexity of given Algorithms.

TEXT/REFERENCE BOOKS

1. Kruse, Tondo and Leung, "**Data Structures and Program Design in C**", 2nd edition, PrenticeHall,1997.
2. Ellis Horowitz, SartajSahni, "**Fundamentals of Data Structures**", Illustrated Edition by Computer Science Press.

24ICxxxT					Fundamentals of ICT					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hrs/Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
2	0	0	2	2	25	50	25	--	--	100

COURSE OBJECTIVES

- Provide students with a comprehensive understanding of the evolution, trends, and components of Information and Communication Technology (ICT).
- Equip students with the necessary skills to effectively manage and evaluate digital information, including information retrieval strategies, digital literacy principles, and data management techniques.
- Familiarize students with modern technologies shaping the future of ICT, such as Artificial Intelligence/Machine Learning, Blockchain Technology, and IoT, and explore their applications across different domains.

UNIT 1: Introduction to ICT	**06 Hrs.**
Overview of ICT: Evolution and Trends in ICT, ICT Infrastructure and Components, Role of ICT in Modern Society, and Industry; Type of ICT Services: Software Development, Computer Networking, Telecommunication, Cybersecurity, Data Analytics, Cloud Computing	
UNIT 2: ICT Technologies	**09 Hrs.**
Hardware Technologies: Data Processing Hardware, Data Input Hardware, Data Output Hardware, Data Transmission Hardware, Data Storage Hardware; Software Technologies: Firmware, Operating Systems, IT Protocols, Programming Languages, Software Development Methodologies; Communication Technologies	
UNIT 3: Information Management and Digital Literacy	**08 Hrs.**
Managing and evaluating digital information, Information retrieval strategies, digital literacy topics such as copyright and plagiarism, data management techniques, and information security practices. digital ethics and security, data management, Modern Technologies for ICT using Artificial Intelligence/Machine Learning, Blockchain Technology, IoT.	
UNIT 4: Emerging Trends in ICT	**05 Hrs.**
Current and future trends in ICT, Application in different domains, futuristic tools and technologies for ICT advancements.	

Total: 28 Hrs

COURSE OUTCOMES

On completion of the course, student will be able to

CO1: Identify various ICT components, and list types of ICT services.

CO2: Demonstrate comprehension of hardware/software technologies, communication technologies, and digital information management principles.

CO3: Apply information retrieval strategies, digital literacy principles, and data management techniques in practical scenarios.

CO4: Analyze the role of modern technologies such as AI/ML, Blockchain Technology, and IoT in ICT.

CO5: Evaluate current and future trends in ICT and assess their applications in different domains

CO6: Design ideas and proposals for futuristic tools and technologies to advance ICT.

TEXT/REFERENCE BOOKS

1. Rajaraman, V. Introduction to information technology. PHI Learning Pvt. Ltd., 2018.
2. Bryant, Randal E., and David Richard O'Hallaron. Computer systems: a programmer's perspective. Prentice Hall, 2011.
3. Schwab, Klaus. The fourth industrial revolution. Crown Currency, 2017.

<Course Code>					Electronic Devices and Circuits					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hrs./Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
2	0	0	2	2	25	50	25	--	--	100

COURSE OBJECTIVES

1. To study the basic principles, configurations and applications of electronic devices
2. To analyze and design transistor amplifiers and practical OPAMP based circuits
3. To understand the operation of D/A & A/D converter types and applications.

UNIT I: SEMICONDUCTOR DIODES AND CIRCUITS	**6 Hrs.**
Review of semiconductor basics, P-N Junction diodes, biasing, reverse saturation current, cut-in voltage, V-I Characteristics. Diode Applications: rectification, half-wave, full-wave and bridge rectifier and design, filters, clippers, clampers. Zener Diodes: voltage regulator, regulator design, varactor diodes, photo diodes, Schottky diodes, tunnel diodes and LEDs.	
UNIT II: Transistor Amplifiers	**8 Hrs.**
Transistors BJT, FET, MOSFET, types, working principal, characteristics, and region of operation, load line analysis, biasing techniques. Transistor as an amplifier, design of a single stage amplifier, gain, bandwidth and frequency response. Frequency response of multistage amplifiers.	
UNIT III: LINEAR INTEGRATED CIRCUITS AND APPLICATIONS	**10Hrs.**
Basic op-amp circuits: Inverting & Non-inverting voltage amplifiers, Voltage follower, Summing, scaling & averaging amplifiers. Linear Applications: Instrumentation Amplifiers, V-I & I-V converters, Differentiators & Integrators, Active filters Non-linear Applications: Precision Rectifiers, Wave Shaping Circuits, Log and Antilog Amplifiers, Comparators and its applications, Sample and Hold circuits. Waveform Generators: Sine, Square and Triangular Wave generators. IC 555 Timer: Monostable operation and its applications – Astable operation and its application	
UNIT IV:	**4 Hrs.**
DAC Specifications: DAC circuits, Weighted Resistor DAC, R-2R Ladder, Monolithic DAC ADC specifications: ADC circuits, Flash Type, Ramp Type, Successive Approximation and Dual Slope ADC, Monolithic ADC.	
TOTAL HOURS: 28 Hrs.	

COURSE OUTCOMES

On completion of the course, student will be able to:

CO1	:	Remember the fundamentals of semiconductor devices
CO2	:	Understand the operation of semiconductor diodes, transistors and operational amplifier
CO3	:	Demonstrate the applications of operational amplifiers
CO4	:	Analyze the ADC and DAC using operational amplifiers
CO5	:	Evaluate performance of transistor circuits and Linear integrated circuits
CO6	:	Design diode rectifiers, voltage regulators, transistor amplifier and linear and nonlinear application of opamp

TEXT/REFERENCE BOOKS:

1. Boylestad and Louis Nashelsky, “Electronic Devices and Circuit Theory”, Eleventh Edition, Pearson India Education, 2015.
2. Sedra, Smith and Chandorkar, “Microelectronic Circuits – Theory and Applications”, Seventh Edition, Oxford University Press, 2017.
3. Donald A Neamen, “Electronic Circuits – Analysis and Design”, Third Edition, McGraw Hill Education, 2006.
4. Ramakant A. Gayakwad, “Op-Amps and Linear Integrated Circuits”, 4th Edition, Prentice Hall, 2000.
5. Roy Choudhury and Shail Jain, “Linear Integrated Circuits”, 2nd Edition, New Age International Publishers, 2003
6. Sergio Franco, “Design with operational amplifier and analog integrated circuits”, McGraw Hill, 1997